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Supporting Online Material for

Tenfold Population Increase in Western Europe at the Neandertal–to–Modern Human Transition

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Summary of analytical methods

Numbers of occupied sites (Fig. 2A, Table 1) were taken from the comprehensive data-base of French Upper Palaeolithic sites compiled by P.-Y. Demars (*41-42)* supplemented for the later Mousterian (MTA) sites by three further sources (*9, 48-49).* 'Sites' were defined as any locations that contained samples of lithic artefacts technologically diagnostic of the three main study periods. *Quantities of stone tools* (Figs 2B, 3A, and Supporting Material 2-5) were obtained from the published site reports, and include (for reasons discussed further below) only formally 'retouched' (i.e. shaped or 'modified') tools. The *quantities of animal remains* in the different sites and periods (Figs 2B, 3B, Tables 6,7) were converted from the published data on Numbers of Identified Specimens ('NISP') and Minimum Numbers of Individuals ('MNI') into the estimated food contributions of the different taxa, as explained further below (Tables 8-10), and entail the assumption that animal food resources (especially those of large ungulates) provided by far the dominant food resource to human groups under the predominantly open, periglacial conditions of both the late Neanderthal and early modern human periods in Europe (*9,29, 44,46 , 50-54*). Data on the overall *areas of occupation* on the sites (Fig. 3C, Table 11, 12) represent in each case essentially *minimum* estimates of the clearly documented extent of the occupation levels for the different periods, as recorded in the available excavation records, and may in several cases represent significant under-estimates of the original extent of the occupation levels in the sites. The 'averages' of these occupation areas shown in Figs. 3C and 4 and Tables 11, 12 should be interpreted in this light.

In calculating the overall 'densities' and 'accumulation rates' of both stone tools and animal food residues the data were standardized for each site in terms of the average numbers of artefacts and faunal remains recorded per square metre of the excavated occupation levels per 1,000 years of the documented occupation sequences (Figs 2. 3, Tables 2-7). (Calculations based on areas of occupation were preferred to those to those based on volumes of sediments, since the latter parameter is heavily influenced by the effects of localized rock falls, or other geological or climatic factors on overall sedimentation rates during the occupation sequences (*55-59*). For individual sites, the overall 'duration of occupation' in the different occupation periods were

based either on radiocarbon dates (where available) or on the assumption that the sites were *potentially* open and available for occupation during the whole of the time ranges known to be spanned by the four main archaeological periods under review (i.e. MTA 'A', MTA 'B', Châtelperronian and Aurignacian) (see Table 1). Clearly, the latter assumption is best supported where (as in most of the sites in our study) the occupation levels in question were both underlain and overlain by occupation levels of closely preceding and ensuing periods of the Palaeolithic succession in the Périgord region (*24, 55, 59*). Further details of all these analyses and associated interpretative issues are given below:

Chronology

The chronology of the different archaeological and demographic periods involved in our study is relatively securely documented for the Châtelperronian and Aurignacian periods on the basis of reasonably abundant and well controlled radiocarbon dates, but more tentative and approximate for the preceding MTA time range, where the available dating (largely beyond the range of reliable radiocarbon measurements) is dependent mainly on less precise measurements from the TL dating of burnt flint samples and the (technically analogous but more procedurally complex) ESR dating of enamel layers in animal teeth (*60*).

For the Châtelperronian, the oldest dates from stratigraphically well documented samples, processed by the most recent pre-treatment and cleaning procedures (to eliminate contamination by more recent, intrusive carbon) centre on ~ 40,500 BP in 'raw' (i.e. uncalibrated) radiocarbon terms, or when calibrated in terms of the recent "IntCal09' calibration curve (*19)*, to ~44,400 cal BP - as documented at the sites of Arcy-sur-Cure (*34)*, Grotte XVI (*61)*, Châtelperron (*62)* and Roc-de-Combe (*58, 61)*.The most recent dates for the Châtelperronian are more potentially problematic, owing to the widely recognized contamination of many of these samples by younger carbon which can (if not fully eliminated by rigorous pre-treatment of bone samples by ultrafiltration or related techniques) produce dates which are spuriously too young by up to several millennia, as for example in many of the samples dated prior to the past 5-10 years, without the use of advanced pre-treatment techniques (*33-34, 63)*. In our estimation, the youngest methodologically and stratigraphically reliable dates for the French Châtelperronian at present

centre on ~ 35-36,000 (uncalibrated) BP, as at the sites of Châtelperron (*62, 64)*, Arcy-sur-Cure (*34)* and Caune de Belvis (*61)*, with a scatter of substantially younger dates measured prior to the development of recent pre-treatment techniques (*62*) The termination of the Châtelperronian is arguably best indicated by the widespread stratigraphic replacement of the Châtelperronian by ensuing Aurignacian industries within the study region from ~ 35,000 radiocarbon years (~ 40,250 cal BP) onwards (see below).

The extent of any chronological and demographic overlap between the final stages of the Châtelperronian and earliest stages of the Aurignacian populations remains at present a matter of debate (*61-62)*. In the Périgord region the oldest well stratified and well dated radiocarbon samples from the earliest, 'classic' Aurignacian levels are those from the basal Aurignacian levels (layer 14) in the long and detailed Aurignacian sequence at the Abri Pataud (*57, 65-66)*, which have recently been dated on several bone samples processed by the newly-developed 'ultrafiltration' techniques (*63)* to ~ 35,000 radiocarbon years BP (~ 40,250 cal BP) (*63*, *67)*. Other early Aurignacian levels with apparently similar dates come from Roc-de-Combe (~ 34,800 BP), La Ferrassie (~35,000 BP) and Abri Caminade (~ 35,400± BP), with a closely similar dating (~ 34,800) for the 'Proto-Aurignacian' level at Arcy-sur-Cure further to the north in Central France (*33-34, 58,61)*. The precise relationship of the 'Proto-Aurignacian' industries and the 'classic' Aurignacian industries remains to be established (*28, 68, 69-70)*, but on the basis of the recent dating at Arcy-sur-Cure, and provisional dates for a similar industry at Saint-Césaire (Charente-Maritime) (*34-67)*, there is no reason to infer a aignificant age difference (if any) between these two industrial variants within the study region (though earlier occurrences of 'Proto-Aurignacian' industries are known from sites in eastern France and the French Pyrenees (*68-70)*. It should be noted that only two of the 17 Aurignacian assemblages included in our study (from Le Piage layer K and Saint-Césaire layer EGF) are attributed to the 'Proto-Aurignacian' group). On this basis, the earliest appearance of Aurignacian/Proto-Aurignacian technologies within the study region can be provisionally dated to ~ 35,000 BP in 'raw' radiocarbon terms, or ~ 40,250 cal BP in calibrated terms.

The end of the Aurignacian sequence, and the interface with the stratigraphically ensuing Gravettian/Upper Périgordian industries, has now been dated in the Abri Pataud sequence to ~

30,500 BP (~ 35,000 cal BP) (*32-33)* with apparently similar dates from the final Aurignacian/Gravettian interface at La Ferrassie (*32-33, 58 ,63, 67)*. The division between the 'earlier' and 'later' Aurignacian is potentially more problematic in both radiocarbon and archaeological terms. In the Abri Pataud sequence there is a well defined transition between the earlier ('Aurignacian I') and later ('Aurignacian II') stages, marked in both the typology and technology of the lithic industries (e.g. increases in the frequencies of nosed scrapers, busqué burins and small Dufour bladelets, and decreases in frequencies of carinated scrapers and Aurignacian blades), and an associated shift from predominantly reindeer to predominantly horse and bovid dominated faunal assemblages (*57,59-60, 71),* dated to ~ 33,000 BP (~ 37,500 cal BP) (*32-33, 63)*. A closely analogous technological and faunal transition in the equally long Aurignacian sequence at La Ferrassie (*56, 58)* appears to have a similar age (*32-33, 58)*. On the basis of these measurements the overall time span of the Aurignacian sequence within the Périgord region would cover a time range of ~ 5,250 years (from ~ 40,250 to 35,000 cal BP), with each of the earlier and later stages of this sequence spanning a period of approximately 2,500-3000 years.

The current dating of the classic sequence of MTA (Type A and B) industries is mainly based on a series of TL measurements of burnt flint sample from the lower shelter of Le Moustier (*31)*, combined with a parallel series of ESR measurements on large animal teeth from both Le Moustier (*32)* and Pech de l'Azé site I (*21)* (with a series of more methodolgically problematic TL dates for the site of Grotte XVI (*72)*). The earliest (Type A) stages of the MTA sequence are dated at Le Moustier to ~55,800$\pm$5000 (TL) (*31)*, to >51,000 BP (ESR) at Pech de l'Azé site I (*21)* and (on more debatable TL measurements) at Grotte XVI to ~ 64,600$\pm$3100 BP (*72)*. On the basis of these dates, the start of the MTA sequence in the study region would seem to date conservatively to at least 55,000 BP, and conceivably as early as ~ 60,000 BP. The transition from the earlier (Type A) to later (Type B) industries is dated at Le Moustier to ~46,300$\pm$3000 BP (TL) (*31),* and at Pech de l'Azé I to ~ 46,000$\pm$6000/51,700$\pm$7000 (ESR) (*21),* suggesting a very approximate age for this transition at around 50,000 BP. The final stages of the MTA sequences are dated at Le Moustier to ~ 42,500$\pm$2000 BP (TL) (*31)* and ~ 41,000$\pm$2600 BP (ESR) (*32)* and at Pech de l'Azé I to ~ 42,000$\pm$7000 BP (ESR) (*21)*. There are five additional radiocarbon-accelerator dates for MTA Type B industries (from the sites of Pech de l'Azé I, La

Rochette and Barbas), all of which fall within the range of ~ 42,000-47,000 BP, in calibrated radiocarbon terms (*21)*. The combined ages for these samples would point to an approximate age for the end of the MTA sequence in the study region of between 43,000 and 47,000 cal BP, which corresponds at least broadly, if not very closely, with the dating of the earliest Châtelperronian levels in the western French region as a whole to ~ 44,000 BP in calibrated radiocarbon terms, as discussed above. The status of certain occurrences of 'Denticulate' and 'Typical Mousterian' industries in relation to those of the MTA group is currently a matter of some debate (*9)*. But on the basis of both the chronometric dating evidence outlined above, and the strong evidence for technological and typological continuities between the final MTA and ensuing Châtelperronian industries (see below), the status of the MTA as a very late episode in the overall Mousterian succession within the study region is not in doubt.

The combined implications of all these dates is that the overall time range of the MTA succession in the southwest French region extends for at least 10,000 years (from at least 55,000 BP to ~ 45,000 cal BP), with individual time ranges for the separate A and B phases of the MTA each spanning periods of at least ca. 5000 years. It should be recalled that if the total time-span of either of these two phases of the MTA exceeded these estimates, then our calculations of the relative 'intensities of occupation' (based on stone tool densities) would reduce to even lower values than those calculated in Tables 2-4, and would correspondingly increase the documented contrasts with the intensity of occupation estimates for the subsequent Aurignacian occupations, with associated demographic implications for changes in Neanderthal versus early modern human population densities discussed in the main text. The same would be true for any suggested increase in the overall time span of the intervening (final Neanderthal) Châtelperronian occupations.

Archaeological/hominin associations

The Neanderthal associations of the French MTA industries would not currently be seriously disputed (*21)*, as an integral part of the broader, European-wide Mousterian/Middle Palaeolithic 'technocomplex', for which explicit Neanderthal associations have now been documented from large numbers of sites across Europe (*73-77)*. In the case of the French MTA, these associations

are best documented at the sites of Pech de l'Azé I (where a well preserved child's skull and mandible excavated in 1909 have recently been attributed on both stratigraphic and archaeological grounds to the later (Type B) stages of the MTA sequence) (*20-21, 78-79)* and (more controversially) in the lower shelter at Le Moustier, where an undisputed Neanderthal skeleton (*76, 80)* was discovered clearly embedded within the later (Type B) MTA deposits on the site, but could possibly be intrusive into these deposits from a closely overlying (conceivably Châtelperronian) level (*20-21, 76, 80-81)*.

The case for the Neanderthal associations of the closely ensuing Châtelperronian industries has been maintained by numerous authors over the past 30 years (*9 , 21,23-24,26, 29,35-36, 39, 74-75,78, 82-93),* on the basis of both direct skeletal associations at the sites of Saint-Cèsaire (with a partial but highly distinctive Neanderthal skull and associated mandible (*37, 88)* and at Arcy-sur-Cure (with a series of 29 distinctively 'archaic' Neanderthal teeth, and a partial temporal fragment (*37, 86, 91-92,94)*, and on the basis of a range of explicit archaeological arguments which link the technological and cultural origins of the Châtelperronian industries with those of the closely preceding later (Type B) MTA industries. As discussed by several authors (*20-22, 25,28, 78, 83, 90, 95-96),* these links are reflected in several different aspects of the Châtelperronian and MTA industries, including the technological and typological similarities between the distinctive 'backed-knife' forms of the MTA industries and the equally distinctive Châtelperron point forms of the Châtelperronian (apparently reflecting mainly a technological shift from flake to blade blanks for the production of the Châtelperronian points) (*9, 20, 22 ,25,83, 95),* the presence in some Châtelperronian assemblages of MTA-like bifacial or small hand-axe forms (*20, 25, 97)* and certain distinctive patterns of core reduction strategies common to both technologies (*20, 25, 83)*. These technological and typological similarities are reinforced by the virtual identity of the documented geographical distributions of the Châtelperronian and MTA industries (both confined to a restricted area of western France, entirely to the west of the Rhône valley (*9, 22, 26)*), and the fact that on both stratigraphic and chronometric dating grounds the MTA represents a very late stage of the overall Mousterian sequence, closely preceding the emergence of the Châtelperronian (*9, 20-21, 25-26 , 95)*. Combined with the anatomical evidence from Saint-Césaire and Arcy-sur-Cure, the demographic and cultural origins of the Châtelperronian within the industries of the MTA group now seems to most recent workers

almost beyond dispute (*9, 74, 21, 23, 25-26,29,35-36,75, 78,83-93),* though an alternative, minority view-point on this issue has recently been discussed (*40)*.

The associations of the classic Aurignacian technologies in Europe with anatomically modern populations has similarly now been almost universally accepted over the course of the past decade (*2-3, 28, 30, 35, 38, 43, 68, 98-101),* following some earlier debate in the literature (*100, 102)*. In the case of the French Aurignacian, these associations are best documented by the indirectly dated skeletal remains from La Quina (*100, 103-104)*, by the archaeological associations of the two modern mandibles from Les Rois (*105)*, and by the series of human teeth from the site of Brassempouy (Pyrenees) which are claimed to show diagnostically modern, as opposed to Neanderthal, morphological patterns (*38, 94, 106).* To judge by the most recent literature (*2-3, 28, 30, 35, 38, 43, 68, 98-101*, the explicitly anatomically-modern human associations of the classic Aurignacian and Proto-Aurignacian technologies would not now be contested by more than at most a very small minority of current workers, though the possibility of some limited interbreeding with the final Neanderthal populations has recently been proposed on the basis of certain anatomical features of some early anatomically-modern remains (*103)* - though such recent interbreeding is apparently not supported in the recent complete genome study of Green *et al*. (*11*).

Database of sites

The following sites were used in the analysis, with information retrieved from the following publications and data sources:

MTA

-Le Moustier (*107*)
-La Rochette (*20, 108*)
-Pech de l'Azé I (*20, 109*)
-Pech de l'Azé IV (*110, 111*)
-Grotte XVI (*20*)
-Laussel (*112*)

Châtelperronian

-Grotte du Renne (Arcy-sur-Cure) (*96, 113*)

-Châtelperron (*64, 114)*

-Saint-Césaire (*115, 116*)

-Roc-de-Combe (*83, 127*)

-La Chèvre (*117*)

-Grotte XVI (*118*)

-Laussel (*112*)

Aurignacian

-Abri Pataud (*66, 71*)

-La Ferrassie (*56, 119*)

-La Chèvre (*117*)

-Caminade Est/Ouest (*55, 120-122*)

-Les Rois (*123*)

-Le Piage (*124-125*)

-Le Flageolet (*119, 126*)

-Roc-de-Combe (*127*)

-Saint Césaire (*116*)

-Grotte XVI (*118, 128*)

-Abri Castanet/Blanchard (personal observation)

-La Quina (*129*)

-Laussel (112)

Numbers of occupied sites

Our data on the relative numbers of occupied sites during the three major chronological and demographic periods of our study were derived for the Châtelperronian and Aurignacian periods from the systematic data-base of French Upper Palaeolithic sites compiled by P-Y. Demars (*42)* supplemented for the MTA sites from two other sources (*48-49)*. For each of these three periods

the site identifications were based on the occurrence of specific, distinctive 'type-fossil' forms, generally recognized as culturally and chronologically diagnostic of the three periods in question - i.e. for the MTA by small, bifacial, 'cordiform' or related 'hand-axe' forms, for the Châtelperronian by typical Châtelperron points, and for the Aurignacian by nosed and carinated scrapers, heavily edge-retouched Aurignacian blades, *busqué* burins and Dufour or Font-Yves bladelet forms (*9, 22 24, 28).* For many of the sites with smaller lithic assemblages unfortunately no clear distinction between the Type A and Type B phases of the MTA, or the earlier and later stages of the Aurignacian, can be reliably made from the available data.

We regard the statistics on the numbers of recorded cave and rock-shelter occupations as inherently much more complete and reliable than those for open-air sites, for a number of reasons. Most of the potential cave and rock-shelter sites for human occupation are generally fairly conspicuous and easily identified features in the present landscape, and have been intensively explored by many generations of archaeologists in the Périgord region over the past 150 years - effectively since the initial pioneering excavations by Lartet and Christy in the 1860s (*24, 59, 130-132).*Although impossible to estimate with any confidence, it is likely that the proportion of these potential sites which have been totally destroyed or obliterated by non archaeological excavations (e.g. road building, house construction, quarrying etc.) in historical times is a relatively limited (though by no means negligible) component of the original site total. A less easily estimated proportion of potential cave and rock shelter sites may of course still await discovery, especially in cases where the roofs or rock overhangs of the sites have subsequently collapsed or been eroded away, or where the deposits have been deeply buried under subsequent rock falls or slope talus deposits. But in any event it is unlikely that any of these factors has any major bearing on the data employed in our study, which is concerned purely with the *relative* frequencies of occupations in different archaeological periods within the documented sample of cave and rock-shelter locations, and not with the *absolute* numbers of occupations which may originally have existed in these sites.

Data on numbers of open-air Middle and Upper Palaeolithic occupations are clearly far less complete and fully documented than those of the cave and rock-shelter sites, for a number of largely self-evident reasons. Most open-air sites are effectively invisible on the surface, and

generally depend for their discovery on chance finds made either in agricultural (mainly ploughing) activities, or other forms of road-building, quarrying or construction works (*132)*. Large areas of the Périgord region are at present heavily forest covered, which effectively precludes the discovery of open-air archaeological sites in non-cultivated areas. Under these circumstances the proportion of the original, existing Middle and Upper Palaeolithic open-air sites in the region which still await discovery is impossible to estimate with any rigour. The only inference which can reasonably be made in this context is that it seems fair to assume that the various factors discussed above will have operated to a roughly equal degree on the sites of all the three major time periods involved in our study (spanning the time range from ~55,000-35,000 years BP) so that the *relative* numbers of open-air sites which have been recorded for the three main demographic periods ought in principle to represent an approximately representative sample of those originally occupied during the individual archaeological periods. The main caveat to be voiced in this context is that open-air occurrences of earlier (Type A) MTA sites are likely to be significantly *over*-represented in comparison to those of Châtelperronian and Aurignacian sites, owing to the much greater visibility and prospects of recovery of the relatively large and visually distinctive biface forms in the course of routine agricultural or other activities, combined with the evident attractions of these pieces to past generations of collectors (*9)*. We see no reason to believe that similar factors should have affected the relative prospects of discovery and identification of Châtelperronian versus Aurignacian sites – i.e. those spanning the critical transition from the Neanderthal to modern human occupations in the region. In this context it may be noted that the recent programme of systematic survey and test excavations conducted along the line of the N89 Autoroute across the northern Périgord region (extending from Bordeaux to Brive) identified a total of six individual Aurignacian sites, compared to only a single Châtelperronian site (*133)* – a very small sample, but one which provides further support for the regional statistics on open-air occurrences documented by Demars (*42)*. Unfortunately only a small handful of open-air Palaeolithic sites have as yet been systematically excavated in the study region (almost all lacking in associated faunal remains) so that we cannot employ data from these sites in our other calculations of occupational areas and intensities of occupation.

On the issue of how far quantitative data on occurrences of Palaeolithic occupations have been affected by subsequent, post-depositional erosion or other site-destruction processes (*134)*,we

again see no major reason to question the reliability of the data we have presented here – at least in the case of occupations recorded in the available sample of cave and rock-shelter sites. The overall time-range of our study (~ 20,000 years) is relatively brief in geological and climatic terms, and we see no clear reason to think that site erosion or destruction factors would have acted to a substantially different degree on occupation levels of the three main demographic periods over this time range - and in particular over the very brief time-period which separated the critical Châtelperronian (final Neanderthal) and early Aurignacian (initial modern human) occupations. In this context it is interesting to note that over the course of the first 20,000 years of the classic Upper Palaeolithic sequence, the total numbers of recorded cave and rock-shelter occupations in the Périgord region actually decreases slightly over the course of time, from a maximum of 108 sites in the Aurignacian to a minimum of ~ 75 sites in the Solutrean period (*42)* These statistics give no reason to suspect a substantial, progressive loss, or selective destruction, of occupation levels in the cave and rock-shelter sites over the particular time-range of our study (*95)*. Surovell *et al*. (*134)* have similarly concluded that progressive erosion or destruction factors are likely to have operated on a much lesser scale in the case of archaeological deposits in cave and rock-shelter sites than on those in more geologically exposed contexts in open-air sites. In the case of our own data-set, it seems highly unlikely that these geological processes would have operated to closely similar degree in the very different geological contexts of both open-air and cave /rock-shelter sites, to yield the closely similar statistics on the relative numbers of Aurignacian versus Châtelperronian occupations documented in both groups of sites (see Fig 2 and Supporting Information Table 1). And of course even if the sites are subject to certain forms of erosion or disturbance, the lithic artefacts themselves are effectively indestructible. Whatever weight may be attached to these potentially progressive site loss or destruction factors, it seems unlikely that this could have had more than a very limited distorting effect on the large-scale contrasts between the overall occupation numbers, occupational-intensities, or occupation-area statistics we have documented over the three main demographic periods of our study. In the same vein we would recall that the broadly similar ratios of Aurignacian versus Châtelperronian occupations documented in the available data from open-air versus cave and rock-shelter sites (with Aurignacian occupations outnumbering Châtelperronian occupations in both groups of sites by a factor of around three or four to one) provides no reason to think that the overall, *relative* use of open-air versus cave/rock-shelter locations differed substantially between these

two (final Neanderthal and early modern human) time ranges. The lack of reliable, systematic data on the numbers of MTA open-air sites (*9)* unfortunately makes it impossible to extend this comparison into the late Mousterian time range.

Intensities of occupation in sites

Estimates of the relative 'intensities of occupation' in the various sites and archaeological periods involved in our study are based on two separate parameters, those of the rates of accumulation of retouched stone tools, and the associated accumulation rates of discarded faunal remains (principally those of large ungulates) as a reflection of the quantities of animal food consumed on the sites. As explained in the main text, both of these parameters are expressed in terms of the rates of accumulation of the occupation residues calculated per square metre of the excavated occupation deposits, per 1,000 years of the estimated occupation sequences,. In principle we would have hoped to incorporate data also on the quantities of un-retouched flakes and associated flaking products in our analyses, but in practice the highly variable size criteria by which these 'waste' products were collected during different excavations and recorded in the published site reports prevented any fully controlled and systematic analysis of these aspects of the lithic assemblages.

The possible variables, and potentially distorting factors, which need to be taken into account in the consideration of these two data-sets can be summarized as follows:

Stone tool densities

The analysis of stone-tool assemblages from European Middle and Upper Palaeolithic sites is by now a fairly routine procedures which rests largely on the long established and widely applied tool typologies of F. Bordes (*135)* for the Middle Palaeolithic industries and D. de Sonneville-Bordes (*24, 136)* for the Upper Palaeolithic industries. In applying these typologies to the various MTA, Châtelperronian and Aurignacian sites in our study sample, to derive relative 'intensity of occupation' values for the different cultural periods, there are a number of potential factors or variables to be kept in mind. These include: potential variations in the total range and relative importance of the different economic and technological activities carried out in the

different sites; the potential effects of variable raw material supplies on the overall rates of tool manufacture, use and discard in the different periods and site locations; similar effects of variable raw material supplies on the rates of reworking, re-sharpening, or recycling of tools in the different site locations; the potential effects of localized variations in the overall spatial distributions of the tools in the excavated occupation levels; and the extent of use of entirely unretouched flakes and blades (as opposed to formally retouched 'tools') in the different periods.

In response to these potential sources of bias or distortion in the data, the following observations can be made:

1. All of the three main demographic groups in our study were occupying essentially the same geographical regions, under a broadly similar (if climatically oscillating) range of climatic and environmental conditions, and exploiting to varying degrees a similar range of animal food supplies. Under these conditions it is reasonable to suggest that the same basic range of material equipment would have been needed to cope with a closely similar range of activities, i.e. hunting equipment, tools for the skinning, butchery and processing of large animal carcases; skin-working equipment (for clothing, blankets, living shelters etc.); and equipment for shaping both wood and (in the case of the Aurignacian and at least some Châtelperronian sites (*34)* but apparently not in the MTA) bone and antler artefacts (*137)*. Use-wear analyses of a range of both Middle and Upper Palaeolithic stone tool forms appear to confirm that these were indeed the principal activities performed by stone tools over the three main chronological periods of our study (*138-141)*. Under these conditions there seems no obvious reason to believe that the overall rates of production, use and discard of stone tools would have varied dramatically over the three major time ranges of our study, at least on a scale to have led to major distortions in our overall comparative statistics for the three periods.

2. The supplies of raw materials for stone tool production can presumably be regarded as an effective constant over the three major time ranges of our study. While the precise qualities and relative accessibility of flint and similar stone for tool manufacture inevitably varied to some degree with the different site locations (*9, 142-143)*, the sites we employed for all the three time periods are distributed over essentially the same range of geological outcrops (i.e. the various

limestone formations of the Aquitaine region), in which all of the cave and rock shelter sites included in our study are located. Accordingly it is unlikely that variations in raw material supplies would have had a large-scale effect on the overall rates of production and discard of stone tools in the different site locations during the different chronological and demographic periods of our study, nor on the rates at which the tools were reworked or re-sharpened during the occupation sequences, especially when the data are 'averaged out' over the substantial numbers of individual sites and individual assemblages on which the data for the different time ranges are based.

3. In the case of the critical comparison between the Châtelperronian (i.e. final Neanderthal) and Aurignacian (initial modern human) periods the tool assemblages are mainly dominated by a broadly similar range of distinctively 'Upper Palaeolithic' tool forms (end scrapers, burins, perforators, truncated pieces and variable retouched blade forms). Together, these forms invariably account for well over 50 percent of the total retouched tool component in both the Aurignacian and Châtelperronian assemblages, and collectively account for the greater part of the documented quantitative comparisons between the two cultural periods (*23, 24, 83,96)*. Clearly this observation cannot be applied to the preceding MTA industries, but in these assemblages again it is reasonable to assume (for the reasons given above) that the same basic repertoire of carcase processing, skin preparation and wood-working activities were carried out with the use of typologically different but functionally analogous stone-tool forms (in the case of the case of the MTA industries principally various forms of side-scrapers, backed knives, notched and denticulated tools and small bifaces) – a conclusion again supported by a number of microscopic use-wear studies of the different tool forms (*138-141)*.

4. The effects of localized variations or concentrations in the overall spatial distribution of stone tools over the excavated occupation surfaces is clearly more difficult to evaluate, in the absence of detailed artefact distribution plots for the majority of the excavated sites. In this context however it should be recalled that most of the individual occupation levels involved in our study were excavated over substantial areas of the sites, usually focused on the most central and heavily occupied areas, and that the great majority if not all of the assemblages clearly represent varying degrees of spatial palimpsests of many separate episodes of human occupation on the

sites (*24, 55)*. Unless closely similar patterns of spatial distributions and localized concentrations of tools were repeated over the whole of these occupation sequences, it seems unlikely that this factor could have had a major impact on the overall quantitative comparisons of tool densities involved in our study.

5. Finally, we would draw attention to the generally close correspondences we have observed between the overall 'intensity of occupation' values derived from the analyses of the stone-tool assemblages and those of the associated animal food residues (as a reflection of food consumed on the sites), over the two most critical time ranges of our study (i.e. Châtelperronian and Aurignacian) for which direct comparisons can be made (unfortunately almost no quantitative data on faunal assemblages are available for the MTA sites in our study). As discussed in the main text, both of these data-sets point to an overall increase in occupational intensity values over the Neanderthal-to-modern human transition by a factor of approximately x1.8 (see Tables 2, 6, 7 and Figs 2-4, of the main text). We see this internal consistency in these two separate and totally independent data-sets as a strong corroboration that both of these potential indices of occupational intensity are indeed telling a closely similar story, which has not been grossly distorted by any of the hypothetical, potential sources of bias or distortion in either of these two separate data-sets.

Faunal densities

Analyses of archaeological faunal assemblages to estimate the total numbers of animals represented in the assemblages, and the potential quantities of food which they could have provided, are potentially susceptible to a number of methodological and interpretative problems which have been widely discussed in the earlier literature (*144-146)*. In principle, these ambiguities can arise from: (a) variable conditions for the survival and preservation of faunal remains in different geological and depositional environments; (b) variable methods and standards for the recovery of faunal remains during excavations; (c) potential destruction of the faunal remains through the actions of both carnivores and various human cultural practices, such as bone-marrow extraction, burning of bones as a source of fuel, or heavy human trampling on the occupation surfaces; (d) variable methods for the analysis and quantification of the faunal remains by different faunal analysts, in terms of both numbers of identified specimens ('NISP')

and minimum numbers of individuals ('MNI') represented in the assemblages; (e) variable spatial patterns of discard of the faunal remains within the sites; (f) the potentially differing intensity of exploitation and processing of the animal carcases for human food – in terms of the relative use of brains, tongues, offal, bone marrow, bone-grease extraction and other culturally specific practices; (g) the effects of the ages and sexes of the animals killed on the quantities of meat provided; and (h) (perhaps most significantly) the effects of the variable body size of the different animal taxa, and the distances from kill sites, on the proportions, and specific body parts of the animal carcases transported from the kill locations into the associated occupation sites.

In response to these potential sources of distortion on estimates of the overall food contributions represented by the recovered faunal assemblages, the following observations can be made:

1. All of the faunal assemblages involved in our study derive from a broadly similar range of depositional and geochemical environments, represented by the highly calcareous limestone deposits in cave and rock-shelter sites. In general these represent very favourable environments for the preservation of faunal remains, and for at least the majority of the sites in our study it seems clear that bone preservation was generally good. Variations in these depositional contexts could potentially arise from differential patterns of *in situ* weathering of the deposits, as a result of variable climatic and associated chemical weathering processes during the depositional sequences. In this context however it should be recalled (as noted above) that each of the principal time-ranges of our study is known to have spanned a number of separate, short-lived climatic oscillations (collectively spanning interstadials GIS 13-7 of the `Greenland ice-core sequence), in a way which is likely to have had broadly similar effects on the overall weathering and faunal preservation conditions during each of the main chronological and demographic periods of our study. Accordingly, there seems no obvious reason to believe that these climatically-related factors would have led to gross differences in overall bone preservation patterns during the three principal chronological periods of our study.

2. Destruction of bones by carnivore action can in principle have a significant effect on bone preservation in certain contexts, especially in the case of faunal assemblages recovered from enclosed cave, as opposed to relatively 'open' rock-shelter sites. With a few exceptions however (notably Roc-de-Combe, Saint-Césaire, Grotte XVI, Châtelperron and Arcy-sure-Cure) all of the faunal assemblages involved in our study derive from relatively open rock-shelter locations, in which both remains of carnivores themselves, and bones demonstrably affected by carnivore action, invariably represent small components of the total faunal assemblages. Interestingly, the faunal density values calculated for the few cave sites in the study sample appear to show no obvious major differences from those recorded for the same periods and time ranges in the contemporaneous rock-shelter sites (see Tables 2, 6, 7), suggesting that even in the case of cave sites, carnivore activity may have been a relatively limited factor affecting the overall accumulation and survival rates of faunal residues.

3. In the same way we see no obvious reason to think that potentially variable methods and standards of recovery of faunal remains during excavation have had a major effect on the composition of the faunal assemblages utilized in our study. While few if any of the excavations involved the systematic use of sieving procedures, it is unlikely that this would have had a major effect on the recovery of the principal articular ends of the bones, and the remains of jaws and teeth of the main ungulate species on which most of the estimates of both NISP and MNI and associated meat weight estimates were based – though this could of course affect the recovery of some of the smaller skeletal elements, such as small, individual incisor teeth and other small elements such as carpals, tarsals and terminal phalanges from the distal limbs. Heavy trampling of the occupation surfaces, and the recovery of bones from heavily compacted occupation deposits, could no doubt have more significant effects on bone recovery, but once again we see no clear reason to think that these factors differed markedly between the different sites and time periods involved in our study – at least when expressed in terms of the overall food-contribution values on which the comparative statistics for the different cultural and time ranges are based (Tables 2, 6, 7).

4. As far as can be assessed from the published faunal reports, there is no reason to suspect that the methods employed for the analysis and quantification of the recovered faunal assemblages

differed substantially between the different sites involved in the study. All of the faunal assemblages were analysed by experienced and respected faunal specialists, who appear to have adopted similar procedures for the identification and quantification of the faunal remains – both in terms of numbers of identified specimens (NISP) and estimates of the minimum numbers of individuals (MNI) represented by the different taxa. With some small differences (e.g. precise numbers of teeth and, in some cases, phalanges and other small extremity bones) all of the six main species of ungulates analysed here have closely similar numbers of potentially identifiable skeletal elements, and the NISP estimates were based routinely on identified articular ends of the bones, on remains of jaws and teeth, or on other readily identified bone 'landmarks'. MNI estimates (when calculated in the published faunal reports) were based routinely on the minimal numbers of the most frequently represented skeletal elements for each of the taxa represented, and excluded estimates based on refitting fragments of long bones or detached antlers or tusk fragments (*147)*.If there were differences in the identification and analytical methods applied to the different faunal assemblages, it seems unlikely that these could account for more than a small part of the overall scale of variation documented between the different sites and chronological periods involved in our study.

5. The method we adopted for the 'extrapolated' estimation of MNI values from numbers of identified specimens (NISP) figures (where no direct MNI estimates were provided in the original faunal reports) require more explanation, and inevitably remain more tentative and approximate. The method we adopted involved a systematic comparison between the reported NISP values and estimated MNI values for the six principal large ungulate taxa involved in our study (bovids, horse, reindeer, red deer. wild boar, rhino) in the available range of faunal data from the five archaeological sites (incorporating six separate faunal assemblages) for which quantitative data on both NISP and MNI counts were given in the published faunal reports. (i.e. for the sites of Abri Pataud, Le Piage, La Chèvre, Saint-Césaire, Grotte du Renne). In choosing the most appropriate 'conversion factor' between these two variables we selected the *median* value of the overall NISP/MNI ratios from the available data-set of six assemblages – principally on the grounds that this value is quantitatively less susceptible to the effects or one or two extreme (i.e. 'outlying') data points than is the calculation of the average (mean) values. The data employed in these calculations are provided in Tables 8 and 9. Interestingly, in the case of

the three most frequent taxa (reindeer, horse and *Bos/Bison*) the calculations yielded broadly similar NISP/MNI conversion ratios of between x13 and x17. The resulting NISP/MNI conversion ratios were then employed in the subsequent calculations of the overall meat-weight contributions of the different taxa for the individual sites and occupation levels shown in Tables 2, 6, 7; see main text Figs. 2-4, for the assemblages - for which no direct MNI estimates were provided in the published faunal reports. In the case of remains of rhinoceros, mammoth and boar it was found that, in instances where MNI values were given in the literature, the MNI estimates of these species were only in one case greater than one individual. On these grounds we felt that it was unjustified to estimate an MNI of more than one animal for these species in any individual faunal assemblage.

6. Meat weight estimates were calculated following the values given by Patou-Mathis (*148)* for the different taxa presented in Table 10 .Several methodological issues are of course inherent in the use of such estimates, especially when they themselves are based on approximate MNI figures. In the majority of cases, information on the ages and sexes of the animals represented were not provided in the published faunal analyses or were difficult to correlate with the published MNI counts. In such cases the meat-yield estimates are based on the adult, and where applicable, male figures, given in Table 10. Thus the figures given should be considered in essence as *maximum* estimates for the potential meat yields utilized in our calculations

7. As noted above, it is less easy to estimate the effects of the ages and sexes of the hunted animals on the associated estimates of overall meat and food contributions, but again there is no obvious indication in the published faunal reports to suggest that these potential variables differed substantially between the different sites and time ranges involved in our study. The same observations apply (with rather more reservations) to the relative use made of various 'non-meaty' parts of the carcases (brains, tongues, offal, bone marrow, bone grease etc.) in the different sites and time periods – though once again we see no clear, *a priori* reason to believe that these factors should have varied very substantially between the different sites and time ranges involved in the study.

8. Probably the main potential source of uncertainty in the faunal calculations derives from variable patterns of transport of different body-parts of the different animal taxa between the actual kill locations of the animals and the associated occupation sites – which has been shown in earlier studies to be heavily dependent on both the relative body-sizes of the animals and the overall transport distances involved (the so-called "Schlepp effect" of Perkins & Daly (*149)* and others). While this factor can have a major impact on the total numbers of skeletal elements (i.e. NISP values) represented by the different taxa in the recovered faunal assemblages, it is generally assumed to have much less effect on estimates of overall MNI values, as represented by the most commonly occurring skeletal elements – though this must inevitably remain a debatable assumption. The procedures we used to 'extrapolate' from NISP to inferred MNI values (where no direct MNI estimates are given in the published faunal reports) are outlined above, and inevitably remain more provisional and approximate.

9. Finally, and most importantly, we would re-emphasize the generally close correspondence we have documented between the overall 'intensity of occupation' values as reflected in the accumulation rates of animal-food residues, and those of the associated stone-tool assemblages, across the two preincipal chronological and archaeological time ranges of our study (i.e. Chatelperrionian and Aurignacian – with no reliable faunal data for the MTA assemblages). As noted above, it is this internal consistency and mutual reinforcement in the patterns reflected in two separate and essentially independent parameters of overall 'occupational intensity' which argues against the impact of any major 'distorting factors' in either of the two separate indices of occupational intensity involved in our study as a whole. In the same vein it should be recalled that the critical comparisons involved in our study between the three major periods of human occupation over the Neanderthal-to-modern human transition are based on the *overall* statistics for each of these periods, derived in each case from a relatively wide range of different sites and separate occupation levels, in which the effects of any localized (i.e. site-to-site or level-to-level) variations in any of the potentially 'distorting' factors are likely to have been minimized and largely 'averaged out', by the inclusion of many different observations and data points in our overall data-base. Our demographic estimates in other words depend entirely on these comparative, *relative* 'intensity of occupation' statistics for the three major time ranges under review, and not on the specifics of the data from just two or three individual sites.

Tables

1. Total numbers of recorded cave/rock shelter and open-air sites in the study region for the three main periods of the Neanderthal-to-modern human transition. Data for MTA sites from (*48*,*49*); data for Châtelperronian and Aurignacian sites from *(42).* For details of age ranges and hominin associations see Supporting Material. Radiocarbon ages are calibrated from (*19*).

Period	Taxon	Age range Kyr BP (approx.)	Cave/rock-shelter sites (N)	Open-air sites (N)
MTA (A/B)	Neanderthal	55.0-44.4	26	No data
Châtelperronian	Neanderthal	44.4-40.25	30	7
Aurignacian	Modern Human	40.25-35.0	108	39

Table 2. Average densities of retouched tools and estimated meat weights in occupation levels of the individual study periods expressed as values per square metre of the excavated deposits per 1000 years of the occupation sequences. As explained in the text the time spans of the different periods are taken as ~ 5000 years each for the MTA A and B phases, ~ 4150 years for the Châtelperronian, and ~ 2500 years each for the earlier and later stages of the Aurignacian (see Table 1). For details of individual sites and occupation levels see Tables 3, 5, 6, 7. and Figs 2,3 (main text) For statistical significant tests for the retouched tools see Table 4. A Mann- Whitney U Test on the differences between the meat weight densities for the Châtelperronian and Aurignacian periods showed this to be statistically significant (p= 0.036).

	Retouched tool densities (N/m^2/1000yrs)			***Meat weight densities (kg/m^2/1000yrs)***		
Period	**N**	**Mean**	**S.E. of mean**	**N**	**Mean**	**S.E. of mean**
MTA A	5	7.6	5.4	No data		
MTA B	4	5.4	1.8			
MTA (total)	9	6.6	3.0			
Châtelperronian	5	9.7	2.4	6	84.6	24.7
Early Aurignacian	6	17.2	3.9	8	163.1	32.3
Late Aurignacian	6	18.1	3.7	7	141.1	27.2
Aurignacian (total)	12	17.6	2.5	15	152.8	20.9

Table 3. Data on retouched tool counts used in the analysis. As explained in the text the time spans of the different periods are taken as ~ 5000 years each for the MTA A and B phases, ~4150 years for the Châtelperronian, and ~2500 years each for the earlier and later stages of the Aurignacian.

Site	Levels	Area excavated (m^2) (approx.)	No. of retouched tools	Tool density (per m^2/1000 yrs)
MTA A				
Le Moustier	G	53	778	2.9
La Rochette	8A,8B	40	315	1.6
Pech de l'Azé 1	4	15	2195	29.3
Pech de l'Azé IV	F4	52	862	3.3
Grotte XVI	C	35	130	0.7
MTA B				
Le Moustier	H	53	1471	5.6
La Rochette	7	13	225	3.5
Pech de l'Azé I	5-7	15	785	10.5
Pech de l'Azé IV	F1-F3	52	554	2.1
Châtelperronian				
Grotte du Renne (Arcy-sur-Cure)	VIII-X	70-80	5252	14.4
Châtelperron	B1-B5	4	181	9.1
Saint -Césaire	EJOP superior	32	305	1.9
Roc-de-Combe	8	12	528	8.8
La Chèvre	1-2	30	945	6.3
Aurignacian				
Abri Pataud	11-14 Early Aurig.	18 (Mean)	1604	35.6
	6-9 Late Aurig.	32 (Mean)	2095	26.2
	Total Aurignacian			30.9
La Ferrassie	K4-K7 Early Aurig.	23	741	12.9
	F1-K3 Late Aurig.	36	2788	31.0
	Total Aurignacian			21.9

La Chèvre	3 Early Aurig.	30	714	9.5
	4 Late Aurig.	30	1273	17.0
	Total Aurignacian			13.2
Caminade-Est	F-G Early Aurig.	38	1100	11.6
	D2s-E Late Aurig.	38	1523	16.0
	Total Aurignacian			13.8
Les Rois	B Early Aurig.	32	1317	16.5
	A Late Aurig.	23 (Mean)	560	9.8
	Total Aurignacian			13.5
Le Piage	F-K Early Aurig.	84	3598	17.1
Le Flagéolet 1	VIII- IX Late Aurig.	50	1049	8.4

Table 4. Results of two-tailed Mann-Whitney U-tests for significance levels of differences in the retouched tool density data across the periods of the Middle-Upper Palaeolithic transition. For the comparisons with the MTA data, the two p-values given relate to comparisons respectively including and excluding the data from Pech de l'Azé I.

	MTA A	MTA B	Châtelperronian	Aurignacian
MTA A	----------	p= 0.327 p=0.083	p=0.310 p=0.063	p=0.062 p= 0.008
MTA B	p=0.327 p=0.083	----------	p=0.190	p=0.014
Châtelperronian	p=0.310 p=0.063	p=0.190	----------	p=0.106
Aurignacian	p=0.062 p=0.008	p=0.014	p=0.106	----------

Table 5. Comparison of tool densities in the Early and Late Aurignacian. See Table 3 for details of individual sites and data sources

Site	Levels	Area excavated (m^2) (approx.)	No. of retouched tools	Tool density (per m^2/1000 yrs)
Early Aurignacian				
Abri Pataud	11-14	18 (Mean)	1604	35.6
La Ferrassie	K7-L1	23	741	12.9
Le Piage	F-K	84	3598	17.1
La Chèvre	3	30	714	9.5
Les Rois	B	32	1317	16.5
Caminade-Est	F-G	38	1100	11.6
Mean				**17.2+ 3.9**
Late Aurignacian				
Abri Pataud	6-9	32	2095	26.2
La Ferrassie	F1-K3	36	2788	31.0
La Chèvre	4	30	1273	17.0
Les Rois	A	23 (Mean)	560	9.8
Le Flagéolet 1	VIII-IX	50	1049	8.4
Caminade-Est	D2s-E	38	1523	16.0
Mean				**18.1+ 3.7**

Table 6. Estimated meat weight densities. As explained in the text the time spans of the different periods are taken as ~ 5000 years each for the MTA A and B phases, ~4150 years for the Châtelperronian, and ~2500 years each for the earlier and later stages of the Aurignacian.

Site	Levels	Area excavated (m^2) (approx.)	Meat weight (kg)	Meat weight (kg /m^2/1000 yrs)
Châtelperronian				
Grotte du Renne (Arcy-sur-Cure)	X, IX	1.5	1136	151.4
Châtelperron	B1-B5	4	2079	103.9
Saint-Césaire	EJOP sup/inf	29	6609	46.0
Roc-de-Combe	8	20	7248	72.4
La Chèvre	1- 2	30	1977	13.2
Grotte XVI	B	50	9210	36.8
Aurignacian				
Abri Pataud	11-14 Early Aurig.	18 (mean)	6795	151.0
	6-9 Late Aurig.	32 (mean)	5089	63.6 .
	Total Aurignacian			107.3
La Ferrassie	K4-K7 Early Aurig.	23	3325	57.8
	F-K3 Late Aurig.	36 (H-K3) 18 (F-G)	15283	219.3 .
	Total Aurignacian			138.6
La Chèvre	3 Early Aurig.	30	3753	50.0
	4 Late Aurig.	30	5840	77.9
	Total Aurignacian			64.0
Les Rois	B Early Aurig.	32	21415	267.7
	A Late Aurig.	23 (Mean)	4304	72.5

	Total Aurignacian			170.0
Roc-de-Combe	7 Early Aurig.	18	7469	166.0
	5-6 Late Aurig.	18	10440	232.0
	Total Aurignacian			199.0
Saint-Césaire	EJO sup, EGF Early Aurig.	17-31	16894	303.6
	EJM, EJJ Late Aurig.	17-22	6435	137.6
	Total Aurignacian			224.7
Le Piage	F-K Early Aurig.	84	41449	197.4
Grotte XVI	Aib, Abb Early Aurig.	90	25070	111.4
Le Flagéolet 1	VIII, IX, XI Late Aurig.	50	23057	184.4

Table 7. Comparison of meat weight densities in the Early and Late Aurignacian. See Table 3 for details of individual sites and data sources

Site	Levels	Area excavated (m^2) (approx.)	Meat Weight (kg)	Meat weight (kg/ m^2/1000 yrs)
Early Aurignacian				
Abri Pataud	11-14	18 (Mean)	6795	151.0
La Ferrassie	K4-K7	23	3325	57.8
Le Piage	F-K	84	41449	197.4
La Chèvre	3	30	3753	50.0
Les Rois	B	32	21415	267.7
Roc-de-Combe	7	18	7469	166.0
Grotte XVI	Aib, Abb	90	25070	111.4
Saint-Césaire	EGF-EJO sup	17-31	1689	303.6
Mean				**163.1+ 32.3**
Late Aurignacian				
Abri Pataud	6-9	32 (Mean)	5089	63.6
La Ferrassie	F-K3	36 (H-K3) 18 (F-G)	15283	219.3
La Chèvre	4	30	5840	77.9
Les Rois	A	23 (Mean)	4304	72.5
Roc-de-Combe	5-6	18	10440	232.0
Le Flagéolet	VIII, IX, XI	50	23057	184.4
Saint-Césaire	EJJ-EGM	17-22	6435	137.6
Mean				**141.1+ 27.2**

Table 8. Breakdown of the meat-weights analysis. NISP counts per taxa per site are show, with the associated MNI figures in parentheses. MNI counts in italics are estimated MNI values based on the methodology outlined in the text (see Table 9). No explicit published NISP counts exist for Les Rois.

		NISP (MNI)							
	Levels	**Bos/Bison**	**Horse**	**Red deer**	**Reindeer**	**Rhino**	**Wild Boar**	**Total estimated Meat weight (kg)**	**Meat weight density (kg/ m^2/ 1000 yrs)**
Châtelperronian									
La Chèvre	1,2	1 (1)	3 (1)	2 (1)	8 (2)	2 (1)	0 (0)	1977	13.2
Saint-Césaire	EJOP sup/inf	137 (6)	79 (4)	11 (1)	129 (5)	7 (1)	2 (1)	6609	46.0
Grotte du Renne	IX, X	4 (*1*)	16 (*1*)	1(*0*)	9 (*3*)	1(*0*)	1(*0*)	1136	151.4
Châtelperron	B1-B5	11 *(1)*	27 *(2)*	1 *(1)*	18 *(1)*	3 *(1)*	0 *(0)*	2079	103.9
Roc-de-Combe	8	102 *(7)*	77 *(6)*	11 *(2)*	112 *(6)*	0*(0)*	2*(1)*	7248	72.4
Grotte XVI	B	26 *(2)*	26*(2)*	122 *(30)*	248 *(14)*	2*(1)*	14 *(1)*	9210	36.8
Aurignacian									
La Chèvre	3-4	120 (8)	71 (6)	13 (2)	298 (19)	1 (1)	3 (1)	9593	64.0
Saint- Césaire	EJJ-EJO sup	337 (15)	609 (17)	18 (5)	4170 (75)	15 (4)	2 (2)	23329	224.7
Le Piage	F-K	441 (29)	242 (24)	14 (4)	3715 (220)	0 (0)	4 (1)	41449	197.3
Abri Pataud	6-9, 11-14	31 (7)	213 (11)	27(6)	823 (51)	0 (0)	4 (2)	11884	107.3
La Ferrassie	F1-K7	209 *(13)*	61 *(4)*	128 *(32)*	240 *(14)*	0 *(0)*	31 *(1)*	18608	138.6
Grotte XVI	Aib-Abb	55 *(4)*	139 *(11)*	339 *(84)*	1122 *(66)*	0 *(0)*	32 *(1)*	25070	111.4
Roc-de-Combe	5-7	69 *(5)*	75 *(5)*	23 *(5)*	2958 *(174)*	3 *(1)*	8 *(1)*	17909	199.0
Le Flagéolet	VIII-IX, XI	113 *(8)*	149 *(11)*	223 *(55)*	1219 *(71)*	0 *(0)*	9 *(1)*	23057	184.4
Les Rois	A-B	(-) 6	(-) 25	(-) 2	(-) 250	(-) 0	(-) 1	25719	170.0

Table 9. NISP and MNI values for different taxa given in the literature from which the utilized conversion ratios were generated. The conversion ratio is the median value of the ratios shown, rounded to the nearest whole number. See note in the text about the usage of wild boar and rhino conversion rates in the analysis. Data from (*71, 116, 117, 125, 150*).

Species	Site	Levels	NISP	MNI	Ratio
Reindeer	Le Piage	F-K	3715	220	16.87
	La Chèvre	3-4	298	19	15.69
	Saint- Césaire	EJOP	129	5	25.80
	Grotte du Renne	VIII-X	2132	18	118.44
	Abri Pataud	6-14	825	51	16.18
	Median				**17**
Bovid	Le Piage	F-K	441	29	15.21
	La Chèvre	3-4	120	8	15.00
	Saint- Césaire	EJOP	137	6	22.83
	Grotte du Renne	VIII-X	65	4	16.25
	Abri Pataud	6-14	31	7	4.43
	Median				**15**
Horse	Le Piage	F-K	242	24	10.08
	La Chèvre	3-4	71	6	11.83
	Saint- Césaire	EJOP	79	4	19.75
	Grotte du Renne	VIII-X	213	16	13.31
	Abri Pataud	6-14	215	11	19.55
	Median				**13**
Red deer	Le Piage	F-K	14	4	3.50
	La Chèvre	3-4	13	2	6.50
	Saint- Césaire	EJOP	11	1	11.00
	Grotte du Renne	VIII-X	2	1	2.00

	Abri Pataud	6-14	27	6	4.50
	Median				**4**
Wild Boar	Le Piage	F-K	4	1	4.00
	La Chèvre	3-4	3	1	3.00
	Grotte du Renne	VIII-X	3	1	3.00
	Abri Pataud	6-14	4	2	2.00
	Median				**3**
Rhino	La Chèvre (Châtelperronian)	1-2	2	1	2.00
	La Chèvre (Aurignacian)	3-4	1	1	1.00
	Saint –Césaire	EJOP	7	1	7.00
	Grotte du Renne	VIII-X	2	1	2.00
	Median				**2**

Table 10. Meat yield estimates used in the calculations of meat weights represented at each site (data from (*148)*) Where the sex or the maturity of the individual was not indicated the estimate for an adult, and (where applicable) male of that species was used. The value given for the horse is the average adult weight given in (*148)* of the species *Equus caballus* and *Equus hydruntius* .

Taxon	Meat yield (kg)
Reindeer	Juvenile: 33 Adult (M): 66 Adult (F): 41
Red deer	Juvenile:110 Adult (M): 187 Adult (F):137.5
Bos/Bison	Juvenile: 550 Adult: 770
Horse	Juvenile: 82.5 Adult:167.75
Wild Boar	Adult: 81
Rhinoceros	Juvenile: 480 Adult: 720

Table 11. Minimum documented areas of occupation in sites of the three main study periods. For details of individual sites see Table 12 and Fig 3c (main text)

Period	N	Mean	S.E. of mean	Documented range (m^2)
MTA	5	110.0	41.4	35-250
Châtelperronian	7	64.7	17.3	20-150
Aurignacian	12	243.8	63.5	18-600

Table 12. Data on minimum areas of documented occupation for MTA, Châtelperronian and Aurignacian occupation levels. The figures represent the minimum areas over which the occupation levels have been shown by excavations to extend. The original, overall occupation areas in many cases could have been substantially larger than the figure shown.

Site	Levels	Minimum documented area of occupation (m^2) (approx.)
MTA		
Grotte XVI	C	35
Pech de l'Azé IV	F1-F4	52
Le Moustier	G,H	53
Pech de l'Azé I	4-7	160
Laussel	9	250
Châtelperronian		
Roc-de-Combe	8	20
Châtelperron	B1-B5	31
Saint Césaire	EJOP superior	32
Grotte XVI	B	50
Grotte du Renne	VIII-X	80
La Chèvre	1-2	90
Laussel	6	150
Aurignacian		
Roc-de- Combe	5-7	18
Les Rois	A-B	32
Le Flagéolet	VIII-IX	50
Le Piage	F-K	84
Grotte XVI	Aib, Abb	90
La Chèvre	3-4	90

Caminade Ouest/Est	All Aurignacian levels	200
La Ferrassie	F1-K7 (Delporte) F-H(Peyrony)	262
Abri Pataud	11-14	500
Abri Castanet/Abri Blanchard	All Aurignacian levels	500
La Quina	All Aurignacian levels	500
Laussel	All Aurignacian levels	600

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